

Finite Element Method

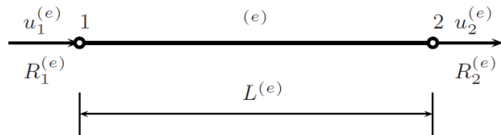
Discrete Systems
MECH4103 — Spring 2026

Dr. Behrouz Arash
Associate Professor in Mechanical Engineering

Oslo Metropolitan University

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- Consider a bar (or spring) element with two nodes, two degrees of freedom, corresponding to two axial displacements. The element supports axial forces only.



- The axial resultant force is obtained by integration of stresses across the thickness direction:

$$N = A^{(e)} \sigma = (EA)^{(e)} \frac{u_2 - u_1}{L^{(e)}} \quad (1)$$

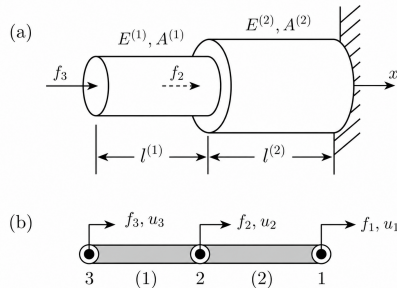
- Taking into account the static equilibrium of the axial forces:

$$R_2^{(e)} = -R_1^{(e)} = N = \left(\frac{EA}{L} \right)^{(e)} (u_2^{(e)} - u_1^{(e)}) \quad (2)$$

- We can write the equations in the form (taking $k^{(e)} = \frac{EA}{L}$):

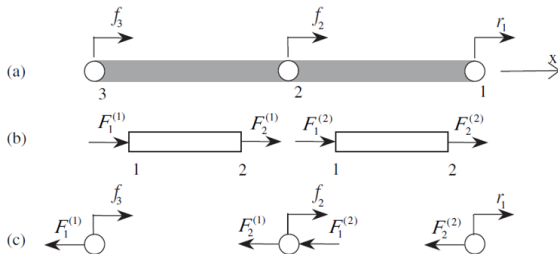
$$\begin{Bmatrix} R_1^{(e)} \\ R_2^{(e)} \end{Bmatrix} = k^{(e)} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1^{(e)} \\ u_2^{(e)} \end{Bmatrix} = \mathbf{K}^{(e)} \mathbf{a}^{(e)} \quad (3)$$

- Consider the two-bar system shown below.



- The first step in applying the FEM is to divide the structure into elements. In the case of a discrete structure, it is necessary only to put nodes wherever loads are applied.
- The finite element mesh consisting of two elements (Figure (b)) is adequate.
- The elements are numbered 1 and 2, and the nodes are numbered 1 to 3; neither the nodes nor the elements need to be numbered in a specific order in FEM.

- For each bar element, the nodal internal forces are related to the nodal displacements.
- To develop the system equations we write the equilibrium equations for the three nodes, constructing free-body diagrams as shown.

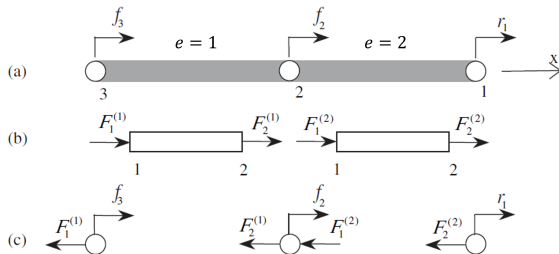


- The system equilibrium takes the form:

$$\underbrace{\begin{bmatrix} 0 \\ F_2^{(1)} \\ F_1^{(1)} \end{bmatrix}}_{\tilde{\mathbf{F}}^{(1)}} + \underbrace{\begin{bmatrix} F_2^{(2)} \\ F_1^{(2)} \\ 0 \end{bmatrix}}_{\tilde{\mathbf{F}}^{(2)}} = \begin{bmatrix} r_1 \\ f_2 \\ f_3 \end{bmatrix} = \mathbf{f} + \mathbf{r} \quad (4)$$

- Each row of the above matrix equation is an equilibrium equation at a node.

- The element forces are labeled with subscripts 1 and 2 (local node numbers).
- The global and local node numbers for this example are shown in Figures (a) and (b).



- For element 1, the global node numbers are 2 and 3, and the stiffness equation gives:

$$\begin{bmatrix} F_1^{(1)} \\ F_2^{(1)} \end{bmatrix} = \begin{bmatrix} k^{(1)} & -k^{(1)} \\ -k^{(1)} & k^{(1)} \end{bmatrix} \begin{bmatrix} u_3 \\ u_2 \end{bmatrix} \quad \left(k^{(e)} = \frac{EA}{L} \right) \quad (5)$$

- Notice that we have replaced the local nodal displacements by the global nodal displacements.

- For element 2 (global nodes 1 and 2):

$$\begin{bmatrix} F_1^{(2)} \\ F_2^{(2)} \end{bmatrix} = \begin{bmatrix} k^{(2)} & -k^{(2)} \\ -k^{(2)} & k^{(2)} \end{bmatrix} \begin{bmatrix} u_2 \\ u_1 \end{bmatrix} \quad (6)$$

- The internal force matrices must be **augmented** (padded with zeros) to match the global size before substitution into Eq. (4):

$$\begin{bmatrix} 0 \\ F_2^{(1)} \\ F_1^{(1)} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & k^{(1)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix}}_{\tilde{\mathbf{K}}^{(1)}} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (7)$$

- Similarly for element 2:

$$\begin{bmatrix} F_2^{(2)} \\ F_1^{(2)} \\ 0 \end{bmatrix} = \underbrace{\begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \\ -k^{(2)} & k^{(2)} & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\tilde{\mathbf{K}}^{(2)}} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (8)$$

- Element 1 exerts no force on node 1 $\Rightarrow$ zero row/column 1 in $\tilde{\mathbf{K}}^{(1)}$.

- Substituting the augmented matrices into Eq. (4), we obtain the **assembled stiffness equations**:

$$\underbrace{\left(\begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \\ -k^{(2)} & k^{(1)} + k^{(2)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix} \right)}_{\mathbf{K}} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ f_2 \\ f_3 \end{bmatrix} \quad (9)$$

- The **assembled stiffness matrix** is:

$$\mathbf{K} = \sum_{e=1}^2 \tilde{\mathbf{K}}^e = \begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \\ -k^{(2)} & k^{(1)} + k^{(2)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix} \quad (10)$$

Note

The stiffness matrix is **singular** ($\det = 0$). To obtain a solvable system, boundary conditions must be prescribed.

- The process of assembling the global stiffness matrix is summarized below.
- **Direct assembly:** bypass zero-padding and add element stiffness terms directly according to their global node numbers.

Matrix scatter and add

Element 1 scatter, global nodes 3 and 2

$$\mathbf{K}^{(1)} = \begin{bmatrix} k^{(1)} & -k^{(1)} \\ -k^{(1)} & k^{(1)} \end{bmatrix} \Rightarrow \tilde{\mathbf{K}}^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & k^{(1)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix}$$

Element 2 scatter, global nodes 2 and 1

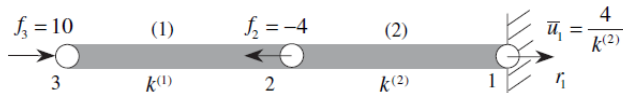
$$\mathbf{K}^{(2)} = \begin{bmatrix} k^{(2)} & -k^{(2)} \\ -k^{(2)} & k^{(2)} \end{bmatrix} \Rightarrow \tilde{\mathbf{K}}^{(2)} = \begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \\ -k^{(2)} & k^{(2)} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Add matrices

$$\mathbf{K} = \sum_{e=1}^2 \tilde{\mathbf{K}}^e = \begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \\ -k^{(2)} & k^{(1)} + k^{(2)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix}$$

Direct assembly

- Consider prescribed displacement $\bar{u}_1 = 4/k^{(2)}$ at node 1 and external forces $f_2 = -4$ and $f_3 = 10$ at nodes 2 and 3.



- Partition the system into **E-nodes** (prescribed displacement) and **F-nodes** (unknown displacement):

$$\begin{bmatrix} \mathbf{K}_E & \mathbf{K}_{EF} \\ \mathbf{K}_{EF}^T & \mathbf{K}_F \end{bmatrix} \begin{bmatrix} \bar{\mathbf{d}}_E \\ \mathbf{d}_F \end{bmatrix} = \begin{bmatrix} \mathbf{r}_E \\ \mathbf{f}_F \end{bmatrix} \quad (11)$$

- Unknowns: $\mathbf{d}_F = [u_2, u_3]^T$
- Solve the F-node equations:

$$\mathbf{d}_F = \mathbf{K}_F^{-1} \left(\mathbf{f}_F - \mathbf{K}_{EF}^T \bar{\mathbf{d}}_E \right) \quad (12)$$

- Reaction force from the E-node equations:

$$\mathbf{r}_E = \mathbf{K}_E \bar{\mathbf{d}}_E + \mathbf{K}_{EF} \mathbf{d}_F \quad (13)$$

- For the two-bar problem, the solution of the unknown displacements gives:

$$\begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} k^{(1)} + k^{(2)} & -k^{(1)} \\ -k^{(1)} & k^{(1)} \end{bmatrix}^{-1} \left\{ \begin{bmatrix} -4 \\ 10 \end{bmatrix} - \begin{bmatrix} -k^{(2)} \\ 0 \end{bmatrix} \left[4/k^{(2)} \right] \right\} \quad (14)$$

which gives:

$$u_2 = \frac{10}{k^{(2)}}, \quad u_3 = 10 \left(\frac{1}{k^{(1)}} + \frac{1}{k^{(2)}} \right) \quad (15)$$

- The reaction force is:

$$r_1 = -6 \quad (16)$$

- Replace the equation corresponding to the prescribed displacement by a trivial equation. Move the first column of $\mathbf{K}$ times $\bar{u}_1$ to the right-hand side and set $u_1 = \bar{u}_1$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & k^{(1)} + k^{(2)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} \bar{u}_1 \\ -4 - (-k^{(2)})\bar{u}_1 \\ 10 - (0)\bar{u}_1 \end{bmatrix} \quad (17)$$

- Solve the full system for all displacements, then recover reactions from the original equations:

$$r_1 = \begin{bmatrix} k^{(2)} & -k^{(2)} & 0 \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ u_2 \\ u_3 \end{bmatrix} = -6 \quad (18)$$

- Impose prescribed displacements by adding a very large number β to the diagonal entry corresponding to the prescribed DOF:

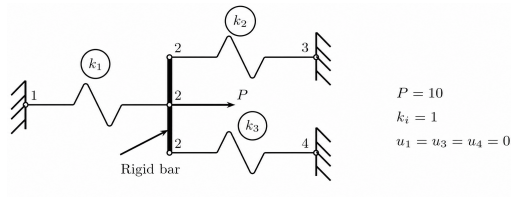
$$\begin{bmatrix} \beta & -k^{(2)} & 0 \\ -k^{(2)} & k^{(1)} + k^{(2)} & -k^{(1)} \\ 0 & -k^{(1)} & k^{(1)} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} \beta \bar{u}_1 \\ -4 \\ 10 \end{bmatrix} \quad (19)$$

- $\beta \sim 10^7 \bar{K}_{ii}$ (average diagonal value) for 8-digit precision.
- The other terms in row 1 and column 1 become negligible, giving equations almost identical to Method 2.

Comparison of Methods

- Method 1 (Partitioning): exact, requires index bookkeeping.
- Method 2 (Elimination): exact, modifies the system directly.
- Method 3 (Penalty): approximate, easy to implement; accuracy depends on choice of β .

- Consider the problem illustrated, where the central bar is **rigid**.



$$P = 10, \quad k_i = 1, \quad u_1 = u_3 = u_4 = 0$$

- Equilibrium equation for each element:

Spring 1:

$$\begin{Bmatrix} R_1^{(1)} \\ R_2^{(1)} \end{Bmatrix} = k^{(1)} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1^{(1)} \\ u_2^{(1)} \end{Bmatrix}$$

Spring 2:

$$\begin{Bmatrix} R_1^{(2)} \\ R_2^{(2)} \end{Bmatrix} = k^{(2)} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1^{(2)} \\ u_2^{(2)} \end{Bmatrix}$$

Spring 3:

$$\begin{Bmatrix} R_1^{(3)} \\ R_2^{(3)} \end{Bmatrix} = k^{(3)} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1^{(3)} \\ u_2^{(3)} \end{Bmatrix}$$

- Compatibility conditions (local $\rightarrow$ global displacements):

$$u_1^{(1)} = u_1, \quad u_2^{(1)} = u_2; \quad u_1^{(2)} = u_2, \quad u_2^{(2)} = u_3; \quad u_1^{(3)} = u_2, \quad u_2^{(3)} = u_4$$

- Node equilibrium ($e = 1, 2, 3$):

$$\text{Node 1: } R_1^{(1)} = F_1$$

$$\text{Node 2: } R_2^{(1)} + R_1^{(2)} + R_1^{(3)} = P$$

$$\text{Node 3: } R_2^{(2)} = F_3 \quad \text{Node 4: } R_2^{(3)} = F_4$$

- Global equilibrium equations:

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1+k_2+k_3 & -k_2 & -k_3 \\ 0 & -k_2 & k_2 & 0 \\ 0 & -k_3 & 0 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ P \\ F_3 \\ F_4 \end{Bmatrix}$$

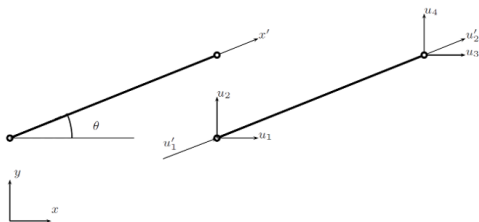
- Applying BCs ($\bar{u}_1 = \bar{u}_3 = \bar{u}_4 = 0$, Method 2):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & k_1+k_2+k_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ P \\ 0 \\ 0 \end{Bmatrix}$$

- Solve u_2 , then recover reactions F_1, F_3, F_4 .

- Consider a typical 2D truss in the global x - y plane. The local axis x' defines local displacements u'_1 and u'_2 :

$$\mathbf{u}'^T = [u'_1 \quad u'_2] \quad (20)$$



- In the global system, the element has four DOFs:

$$\mathbf{u}^T = [u_1 \quad u_2 \quad u_3 \quad u_4] \quad (21)$$

- Transformation between local and global displacements (θ = angle between x' and x):

$$\begin{aligned} u'_1 &= u_1 \cos \theta + u_2 \sin \theta \\ u'_2 &= u_3 \cos \theta + u_4 \sin \theta \end{aligned} \quad (22)$$

or in matrix form $\mathbf{u}' = \mathbf{L}\mathbf{u}$, with:

$$\mathbf{L} = \begin{bmatrix} l & m & 0 & 0 \\ 0 & 0 & l & m \end{bmatrix} \quad (23)$$

- The direction cosines l and m are obtained from nodal coordinates:

$$l = \frac{x_2 - x_1}{L_e}, \quad m = \frac{y_2 - y_1}{L_e} \quad (24)$$

where

$$L_e = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (25)$$

- Local stiffness matrix (same as bar element):

$$\mathbf{K}' = \frac{EA}{L_e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (26)$$

- Strain energy in local coordinates: $U_e = \frac{1}{2} \mathbf{u}'^T \mathbf{K}' \mathbf{u}'$
- Substituting $\mathbf{u}' = \mathbf{L} \mathbf{u}$: $U_e = \frac{1}{2} \mathbf{u}^T [\mathbf{L}^T \mathbf{K}' \mathbf{L}] \mathbf{u}$
- Global stiffness matrix:** $\mathbf{K} = \mathbf{L}^T \mathbf{K}' \mathbf{L}$, or:

$$\mathbf{K} = \frac{EA}{L_e} \begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \quad (27)$$

- Taking into account the definition of strain in the bar, the axial stress is:

$$\sigma = E \frac{u'_2 - u'_1}{L_e} = \frac{E}{L_e} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{Bmatrix} u'_1 \\ u'_2 \end{Bmatrix} = \frac{E}{L_e} \begin{bmatrix} -1 & 1 \end{bmatrix} \mathbf{u}' \quad (28)$$

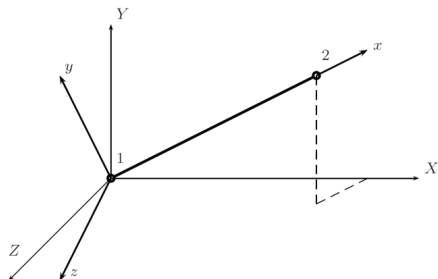
- By transformation to global coordinates ($\mathbf{u}' = \mathbf{L}\mathbf{u}$):

$$\sigma = \frac{E}{L_e} \begin{bmatrix} -1 & 1 \end{bmatrix} \mathbf{L}\mathbf{u} = \frac{E}{L_e} \begin{bmatrix} -l & -m & l & m \end{bmatrix} \mathbf{u} \quad (29)$$

Summary

Once the global nodal displacements $\mathbf{u}$ are known, element stresses are recovered by the above formula using the direction cosines l, m of each element.

- A typical two-noded 3D truss element is illustrated below.
- Each node has **three** global degrees of freedom (u_X, u_Y, u_Z).



- Global stiffness matrix in 3D (direction cosines C_x, C_y, C_z):

$$K = \frac{EA}{L} \begin{bmatrix} C_x^2 & C_x C_y & C_x C_z & -C_x^2 & -C_x C_y & -C_x C_z \\ & C_y^2 & C_y C_z & -C_x C_y & -C_y^2 & -C_y C_z \\ & & C_z^2 & -C_x C_z & -C_y C_z & -C_z^2 \\ & \text{sym} & & C_x^2 & C_x C_y & C_x C_z \\ & & & & C_y^2 & C_y C_z \\ & & & & & C_z^2 \end{bmatrix} \quad (30)$$

- Direction cosines from nodal coordinates:

$$C_x = \frac{x_2 - x_1}{L}, \quad C_y = \frac{y_2 - y_1}{L}, \quad C_z = \frac{z_2 - z_1}{L} \quad (31)$$

Next: Strong and Weak Forms — 1D Problems

Thanks for your attention!